**Third System**
AI Presence Index

# The Structure That Wasn't There

## Cross-model consistency does not inherit the brand-presence hierarchy — and the way it fails tells us exactly what to build next.

June 2026 · Pablo Ulpiano González Castro

## AIAS™ Measurement Program · Phase v0.36

# Before any data ran, this phase pre-registered a specific bet: that “AI consistency” clusters would turn out to be brand presence wearing a different name. The bet lost — and not because consistency revealed a hidden order of its own. The measuring instrument broke first, in three instructive ways. This report scores the loss as registered, then reads the wreckage.

**P1** AI consistency, as currently measurable, is not a hidden brand ranking: consistency clusters do not recover the presence hierarchy.
**P2** The instrument fails where brands are weakest: a computability floor silently erased 15 of 24 premium-spirits brands before the test could run.
**P3** Saturated categories are noisier, not quieter: recognition ceilings appear to amplify cross-model dispersion rather than collapse it — the pre-registered prediction reversed.
**P4** A faint cross-category signature exists in how models disagree: three clusters, one hair under the evidentiary bar. Suggestive, not decision-grade.
**P5** Every failure points one direction: consistency needs a mean-independent instrument before it can carry managerial weight.

This phase asked whether cross-platform consistency (CPC) — how uniformly six frontier models recall a brand — has regime structure of its own, or merely inherits the brand-presence hierarchy already measured by the AIAS™ Presence component. The design was fully retrospective: 112 brand-units across five categories, frozen data, zero new model calls. Decision rules, thresholds, and predictions were locked and deposited before any data was analyzed; the build-time corrections that followed were authored and timestamped outcome-blind, before a single statistic existed.

The pre-registered prediction was “borrowed structure”: consistency clusters would recover presence regimes, and would dissolve once presence was statistically removed. The result landed in the opposite cell of the locked verdict matrix — unstructured. The headline prediction was wrong, and is reported as wrong; the one call that held was the narrower one, that consistency carries no presence-independent structure.

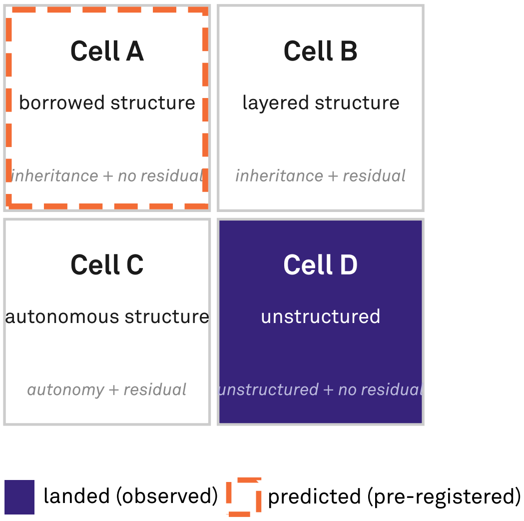
But the three individual failures share an anatomy. The inheritance test couldn’t run at power because the consistency metric is mathematically undefined for low-recall brands — a coverage hole, not a verdict. The saturation test reversed direction because the metric’s dispersion is driven by the recall mean it divides by. And the strongest structural signal in the data appeared only in a sensitivity arm that removed the mean entirely. Three failures, one diagnosis: the current consistency metric is welded to presence by construction.

That diagnosis was already suspected — four prior phases flagged it. This phase adds the regime evidence and converts the case for a mean-independent consistency instrument (methodology phase v1.8) from motivated to overdetermined.

## Verdict matrix — predicted Cell A, landed Cell D

pre-registered prediction falsified; reported as-scored.

*Inheritance and saturation both fail (saturation reverses); residual-structure null as predicted.*




### Verdicts

**Inheritance**
NOT SUPPORTED
*scope: v0.23 (n=9)*

**Autonomy**
NOT SUPPORTED
*scope: omnibus + v0.23 ARI*

**Residual structure**
NOT SUPPORTED
*scope: v0.23 primary*

**Saturation degeneracy**
NOT SUPPORTED
*scope: omnibus (112)*

*Net: Cell D (unstructured). Predicted Cell A (borrowed structure) — falsified.*

Source: AIAS™ v0.36 (CPC Regime Emergence) | Protocol v0.36-prereg-r2 | Third System™

The locked verdict matrix. The phase pre-registered the top-left cell - borrowed structure - and landed in the opposite corner: unstructured. Inheritance and saturation both miss; the residual-independence call was the one held prediction.

## What we measured

v0.36 is a re-analysis: no new AI queries, no new brands. The inputs are the program's frozen omnibus panels measured across the same six-model panel used since v0.17 — audiophile headphones, skincare, cosmetics, automotive, and premium spirits: 112 brand-units in all. CPC is how uniformly the six models recall a brand. Every clustering parameter — the algorithm, the cluster-count selection rule, the thresholds, the permutation tests, the random seed — was locked at pre-registration and deposited to OSF; the seed-bearing scoring code was committed under tag v0.36-prereg-r2, and the deposited amendment record attests that no statistic had been computed when it was written.

The regime target — the per-brand presence classification (Dominant / Established / Emerging / Absent) — is frozen in the premium-spirits phase, the only category where such labels exist. The pre-registration documents that constraint rather than papering over it.

Two principal corrections were made during the build and timestamped before any result existed: a data-lineage error (the per-brand regime labels are materialized in one category, not all five) and a construct-naming collision (the regime target is the premium-spirits presence-quartile classification, not the program's substrate-level Trends taxonomy that happens to share the word "regime"). Scope and analysis-range clarifications were logged alongside. All are public in the deposit record.

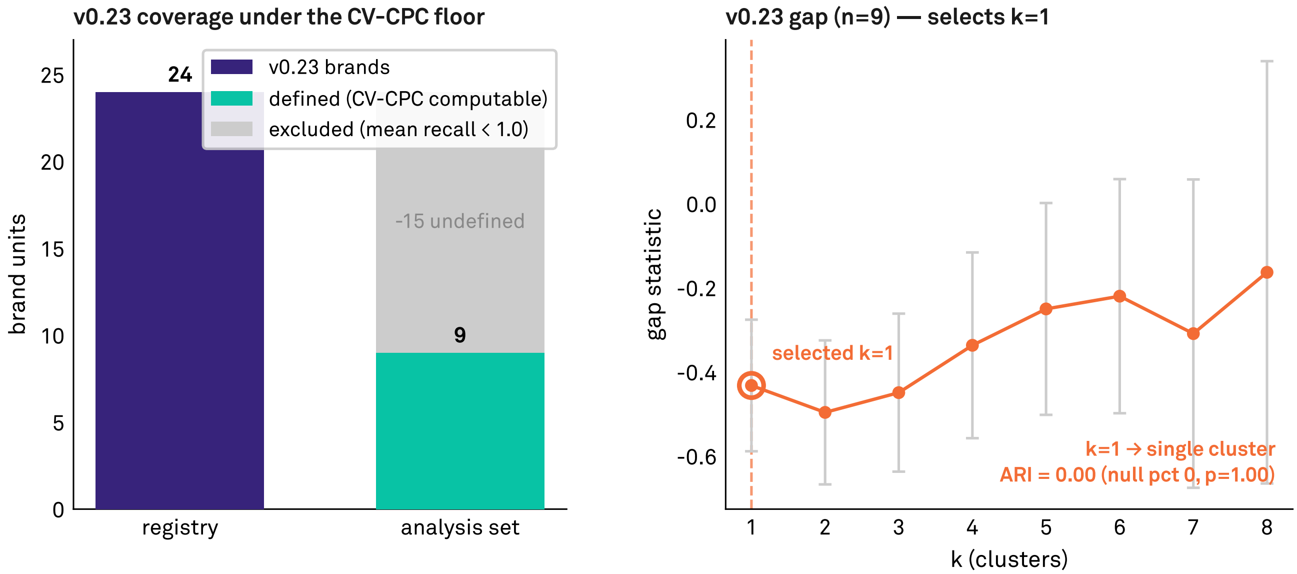
PROPOSITION P1

# AI consistency is not a hidden brand ranking

### Inheritance is unevaluable at power — the floor leaves n=9

The CV-CPC computability floor reduces v0.23 from 24 brands to 9; gap then selects k=1.

*With one cluster the Adjusted Rand Index against the Presence-quartile regime is 0 by construction.*




*Inheritance NOT SUPPORTED — computability floor; test underpowered (n=9), recorded as-scored.*

Source: AIAS™ v0.36 (CPC Regime Emergence) | Protocol v0.36-prereg-r2 | Third System™

**Where the test could run, it ran on nine brands - the computability floor erased the other fifteen - and a single forced cluster left zero agreement with the presence regime. The instrument, not the brands, set the limit.**

The bet was that consistency clusters would recover the brand-presence hierarchy — that a brand's consistency rank would track its presence rank. It does not. Where the test could be run, cluster structure and the presence-regime labels share no measurable agreement (Adjusted Rand Index 0.00). On the locked verdict matrix the phase landed in the opposite corner from the prediction: not “borrowed structure” but “unstructured.”

That is the headline, and it is reported exactly as it scored. What the test could actually see, though, is the real story — and it is smaller than it should have been (see P2).

PROPOSITION P2

# The instrument fails where brands are weakest

Consistency is a ratio that divides by how often a brand is recalled. Below a floor of recall, the metric does not exist — you cannot measure the consistency of something that is never named. In premium spirits that floor erased 15 of 24 brands, leaving 9 measurable. These are exactly the weak-recall brands a consistency diagnostic should care about most.

A metric blind to the bottom of the market cannot classify the market. The inheritance test above ran on those nine survivors — too few, too top-heavy, for a verdict to mean much. We record it as scored and name the limitation plainly.

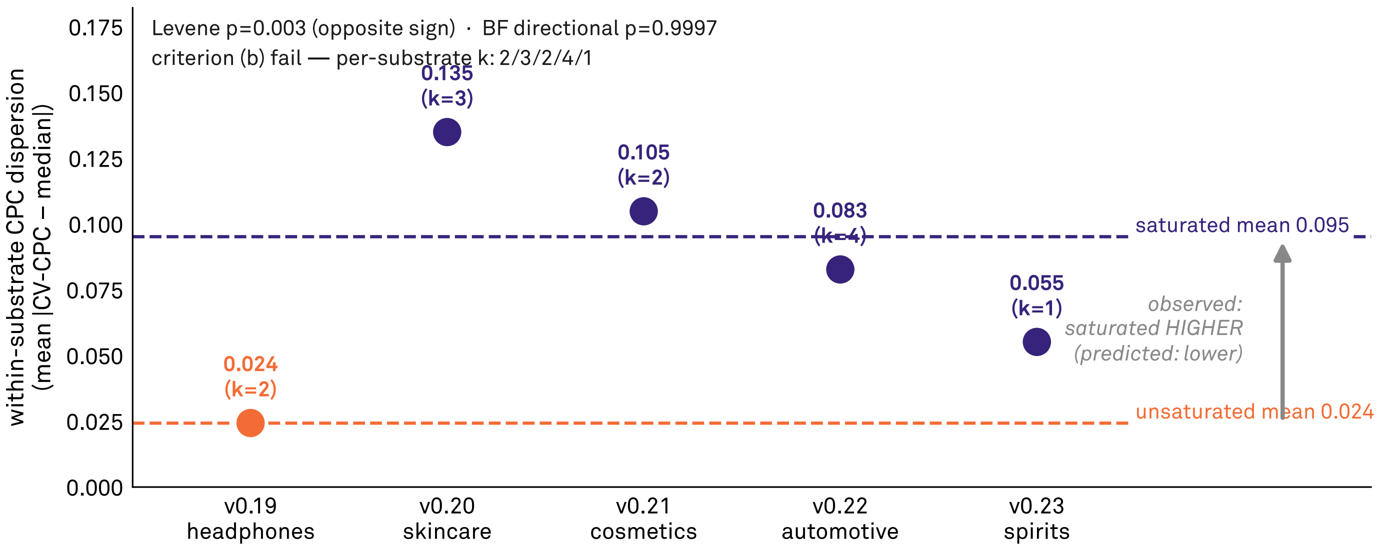
PROPOSITION P3

# Saturated categories are noisier, not quieter

### Saturation reversal — saturated substrates carry MORE CPC dispersion

Within-substrate CPC dispersion by substrate; indigo = saturated (v0.34), copper = unsaturated.

*The pre-registered prediction (saturated collapse, lower dispersion) is reversed; criterion (b) also fails.*




*Saturation degeneracy NOT SUPPORTED — direction reversed; variance higher where recognition saturates.*

Source: AIAS™ v0.36 (CPC Regime Emergence) | Protocol v0.36-prereg-r2 | Third System™

Saturated categories carry MORE cross-model dispersion, not less; the dashed means and the arrow mark the pre-registered direction reversing. Read as post-hoc mechanism - a ratio metric whose denominator shrinks in low-recall tails - not as a finding.

The prediction was that categories where every brand is well recognized would show flatter, more uniform consistency — a ceiling effect collapsing the variation. The opposite happened. Saturated categories carry more cross-model dispersion, not less (0.095 versus 0.024), and the difference is significant in the wrong direction (Levene p=0.003).

Read post-hoc, this is mechanically consistent with a ratio metric whose denominator shrinks across long, low-recall tails — the same presence-coupling that undoes the other tests. Offered as interpretation, not as a pre-registered finding.

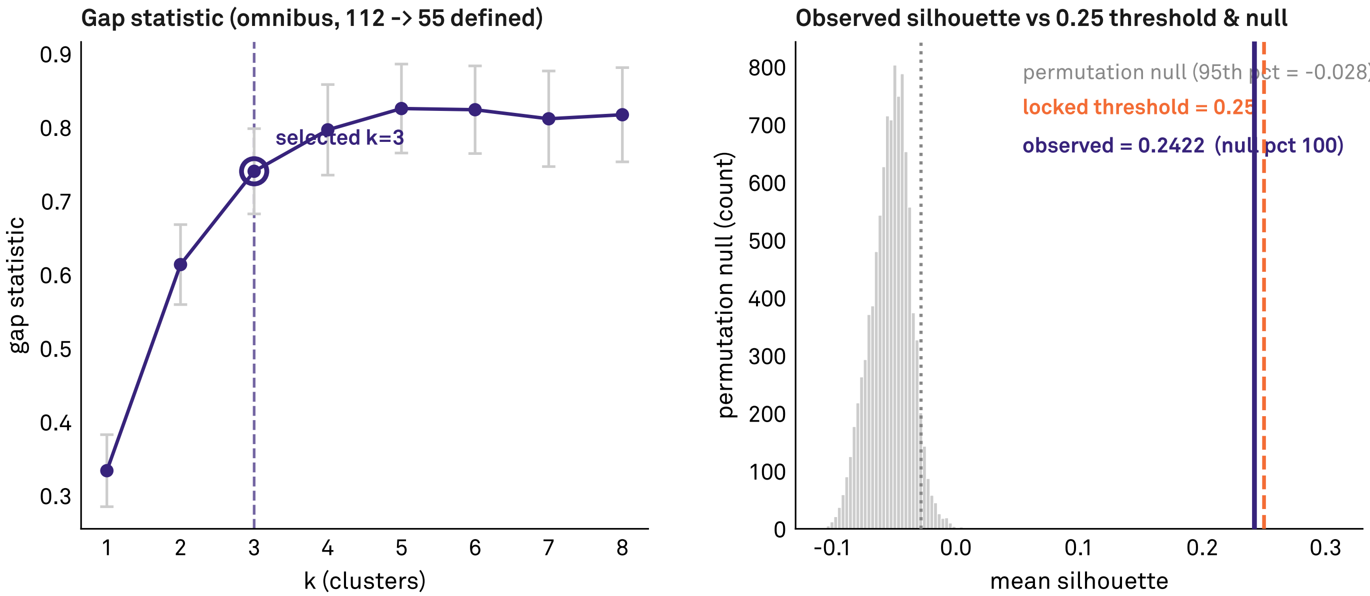
PROPOSITION P4

# A faint signature exists in how models disagree

## Omnibus CPC structure is weak — above null, below the locked bar

Ward/Euclidean clustering of the 6-dim per-model CPC signal; gap selects k=3.

*Silhouette 0.2422 exceeds the 95th-percentile null but falls under the 0.25 autonomy threshold — a near miss.*




*Autonomy NOT SUPPORTED — silhouette below locked 0.25 threshold (though above the permutation null).*

Source: AIAS™ v0.36 (CPC Regime Emergence) | Protocol v0.36-prereg-r2 | Third System™

Pooled across 112 brand-units, three clusters rise above the permutation null - but the silhouette sits a hair under the locked 0.25 bar. Real enough that the next instrument must explain it; not strong enough to score.

Pool all 112 brand-units and a structure does flicker into view: three clusters emerge above chance. But it sits at silhouette 0.2422, a hair under the locked evidentiary bar of 0.25 — above the noise floor, below the line we set in advance. We score it not supported and relax no threshold to rescue it.

It is described here because it is real enough that the next instrument must explain it: there is something in how the models disagree about lesser-known brands, even if today's metric cannot resolve it cleanly.

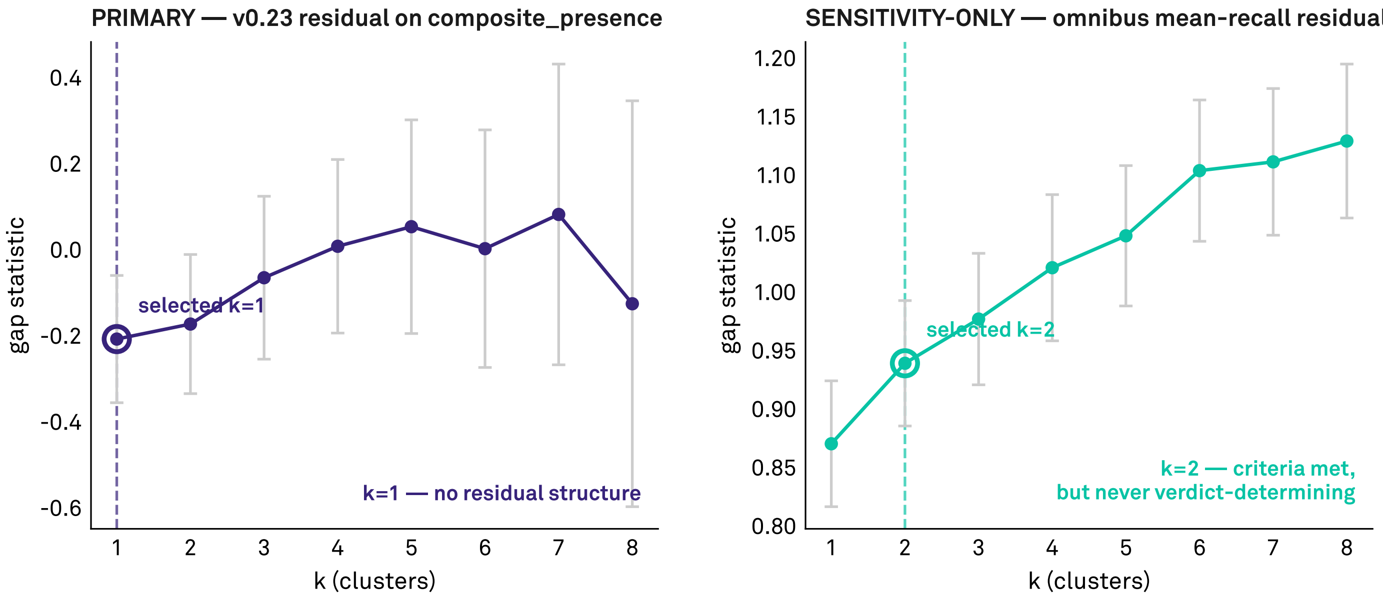
PROPOSITION P5

# Every failure points the same direction

### Residual structure: primary says no, sensitivity arm diverges

Left: locked primary (v0.23, residualized on composite_presence). Right: omnibus mean-recall residual arm.

*The verdict rests on the primary (k=1, no structure); the sensitivity arm (k=2) is reported but not verdict-determining.*




*Residual structure NOT SUPPORTED — primary arm (v0.23), as predicted.*

Source: AIAS™ v0.36 (CPC Regime Emergence) | Protocol v0.36-prereg-r2 | Third System™

Two gap curves. The locked primary (left) finds no residual structure; the sensitivity-only arm that removes recall volume entirely (right) does. The one place structure survives is the one place the mean is gone.

One arm of the analysis removed recall volume from the signal entirely — and only there did structure appear (two clean clusters, criteria met). It was pre-specified as a sensitivity check, never able to set the verdict, and it is treated that way. But the tell is unmistakable: the one place structure survives is the one place the mean is gone.

Inheritance unevaluable, saturation reversed, structure only without the mean — three different failures, one diagnosis. The consistency metric is welded to presence by construction. That is the specification for what comes next.

# Limitations

The inheritance and residual tests ran on a single category at nine defined units — unevaluable at meaningful power, recorded as-scored rather than adjudicated. The regime target exists for one substrate only; the omnibus version is blocked by a binary-versus-graded recognition asymmetry across the earlier phases, which measured recognition as a yes/no rather than a graded signal. The mean-recall sensitivity arm was pre-specified as never verdict-determining and is treated accordingly. And the consistency quantity here holds characterization status only — it was never adopted as a canonical instrument, which is, in the end, part of what this phase is the case for.

# What's next

Three threads follow. First, the v1.8 mean-independent consistency instrument, now carrying five convergent lines of motivation — the v1.7 presence-coupling result and the saturation-collapse findings of v0.33, v0.34, and v0.35, joined now by this phase's regime evidence — with the regime question designated for re-test once the instrument exists. Second, a common-protocol graded-recognition re-scoring of all substrates, which would unlock the omnibus regime target this phase could not assemble. Third, the substrate-level Trends-based taxonomy test, which requires new external data and is explicitly out of retrospective scope.

# The hypotheses, scored

Each row is a locked pre-registered test, scored against its frozen threshold. Plain names stand in for the formal hypothesis labels; the net cell is the verdict-matrix landing.

| Hypothesis | Scope | Predicted | Verdict |
|---|---|---|---|
| **Inheritance** | spirits, n=9 defined | *Supported* | Not Supported — unevaluable at power |
| **Autonomy** | omnibus 112 + spirits ARI | — | Not Supported — silhouette 0.2422 < 0.25 |
| **Residual structure** | spirits primary | *Not supported* | Not Supported — sensitivity arm diverges |
| **Saturation degeneracy** | omnibus 112 | *Supported* | Not Supported — direction reversed |
| **Net** | verdict matrix | *Cell A (borrowed structure)* | **Cell D — unstructured** |

## The numbers behind the verdicts

For readers checking the work against the deposited record (osf.io/ec6wh/v36). All statistics are reproducible from the locked scorer at seed 36.

**Inheritance.** Gap selection chose k=1 on the nine defined premium-spirits units — a single cluster — so the Adjusted Rand Index against the presence-quartile regime is 0.00 (null percentile 0, p=1.00). With one forced cluster, zero agreement is mechanical, not evidential.

**Autonomy.** Across all 112 units gap selected k=3; mean silhouette 0.2422 clears the permutation null (null percentile 100) but falls under the locked 0.25 bar. Above chance, below the line — not supported, no threshold relaxed.

**Residual structure.** Residualized on the presence composite, premium spirits resolved to k=1 (no structure). The omnibus mean-recall sensitivity arm resolved to k=2 and met the internal-validity criteria — but it is sensitivity-only and never verdict-determining.

**Saturation degeneracy.** Within-substrate dispersion ran higher in saturated categories (0.095) than unsaturated (0.024); Levene p=0.003 with the directional test at p=0.9997 (wrong sign). Per-substrate cluster resolution 2/3/2/4/1 also fails criterion (b). Both legs miss; the prediction reversed.

A measurement program proves itself on the misses. The prediction was locked, timestamped, deposited, and wrong — and the falsification did more work than a confirmation would have: it located the failure inside the instrument rather than the brands, and specified the repair. Consistency remains a component worth building. v1.8 is where it gets an instrument that can carry it.

# About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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### Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.7). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

### Citation

González Castro, P. U. (2026). ***Does Cross-Platform Consistency Have Regime Structure? A Pre-Registered Clustering Test Against a Frozen Brand-Presence Classification (AIAS CPC, v0.36)***. Third System. Pre-registration osf.io/ec6wh/v36 (tag v0.36-prereg-r2).

### Underlying datasets

Verdicts: osf.io/ec6wh/v36/scoring/v36_verdicts.json

Clustering intermediates: osf.io/ec6wh/v36/scoring/data/

Scoring code: osf.io/ec6wh/v36/ (scripts/score_v0_36.py)

Pre-registration: osf.io/ec6wh/v36/prereg/ (tag v0.36-prereg-r2)

**Methodology log:** v0.36 (re-analysis; CPC characterization quantity, v1.7).

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